

Prescribed-front Icepack experiment at EKaS: implementation and validation status

Icepack-Sediment

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Result status. The implemented SSA solver has executed paired, short synthetic fixed-front and advancing-front simulations. Their initial and post-spinup states agree, the front follows its prescribed offsets, thickness stays positive, and the discrete mass budget closes to floating-point precision. A calibrated historical EKaS simulation has not been executed. No agreement with Dachauer et al. is claimed. The requested OOI-Image template was not found; its exact AGENTS.md and plotting style remain pending. Figure formatting is provisional.

1 Scientific comparison

Dachauer et al. (2026), doi:[10.1017/jog.2026.10172](https://doi.org/10.1017/jog.2026.10172), Figure 13, examines how prescribed front advance changes flux and thickness at a gate approximately 1 km upstream of the 1953 front. This implementation compares a fixed-front control with a prescribed advancing front over 1953–2021, after a common 25-year relaxation. The initialization is nominally 2021 geometry truncated at the 1953 terminus; it is not a measured 1953 ice surface. The publication’s inversion, two-year relaxation and re-inversion are not reproduced. The approximate offsets used here are 0, 960, 1400 and 1820 m in 1953, 1985, 2007 and 2021. These integrate rounded Figure 13 rates rather than digitized front coordinates. Monotone PCHIP provides smooth position; piecewise-linear interpolation is also available to match the paper’s interpolation convention.

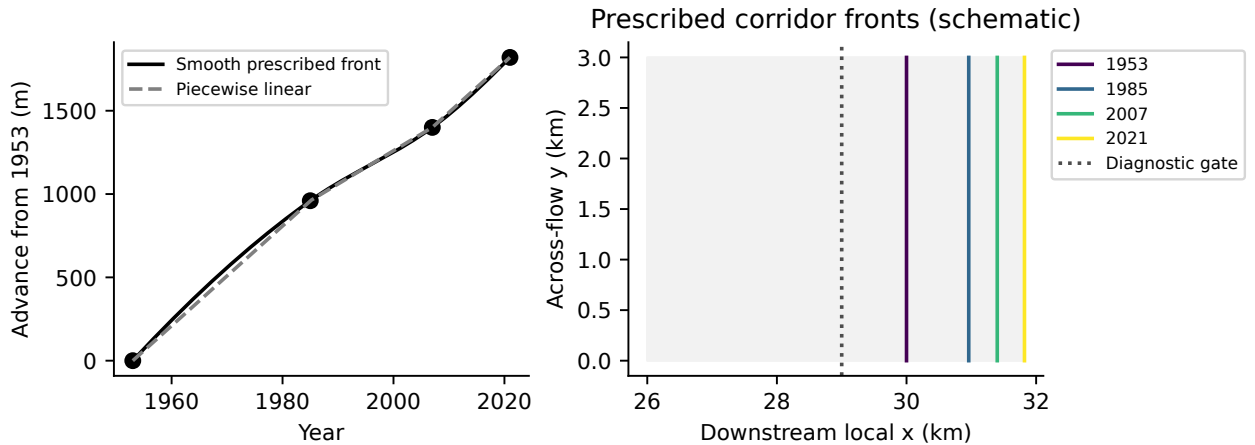


Figure 1: Prescribed approximate chronology and schematic corridor-front positions. These are model coordinates, not a georeferenced reconstruction of observed front lines.

2 Physics and domain motion

Icepack’s `IceStream` supplies SSA stress balance, Glen exponent $n = 3$, ocean traction and the hydrostatic surface convention. A prepared MAR 2021 annual SMB field is held fixed in both experiments. BedMachine Greenland v5 is the stock bed/surface source; a supplied corrected bed can replace it. The authors’ exact kriged bed and inverted fields were not obtained. The model uses a straight 30 km by 3 km trunk corridor with observed inflow; this differs from the paper’s full ice-divide-bounded basin and curved fronts.

With speed v , basal coefficient $k = C_{\text{Ua}}^{-1/3}$ and effective pressure N , the implemented sliding stress approaches

$$t_b = \frac{\mu N k v^{1/3}}{\mu N + k v^{1/3}}, \quad N = \max\{\rho_i g H - \rho_w g \max(-z_b, 0), 0\}.$$

Here $z_b = s - H$ is the ice base, including floating geometry. Floating ice has zero basal traction. Icepack expects a dissipation potential rather than a traction expression, so a 16-point quadrature integrates this stress over speed. A speed regularization of 0.001 m/year avoids the zero-speed singularity. The default $\mu = 0.5$ and uniform $A = 10$ in MPa–m–year units are configurable assumptions, not recovered Ua parameters. A provided friction raster or a Weertman driving-stress prior is used without a parameter inversion.

Each timestep creates a new triangular corridor mesh of prescribed length. Identical topology and node ordering allow thickness transfer with the old/new length ratio, preserving total ice volume. Implicit continuity transport uses $\mathbf{u} - \mathbf{w}$, physical ice velocity minus mesh velocity. The split update is first-order and must be checked for temporal and spatial convergence. Boundary mass flux is evaluated relative to the moving boundary; any inflow uses the supplied initial thickness. No calving prediction or ad hoc new ice slab is imposed. CG1 diagnostics use the actual element connectivity. Nonpositive or nonfinite thickness aborts a run rather than being clipped.

3 Executed checks

The available local Docker image `vdv-ice-shelf` contains working Firedrake and UW Icepack. A synthetic bed, surface and zero SMB were used with a 12 by 4 mesh, $\Delta t = 0.1$ year, a 0.1-year spinup and 0.2-year post-spinup transient. This is a numerical test rather than a regional climate experiment. The scientific default remains a 25-year relaxation and a 68-year transient.

Check	Fixed front	Advancing front
Minimum thickness (m)	456.090	455.807
Maximum relative mass residual	$1.33e - 16$	$1.15e - 16$
Final front offset (m)	0	7.178

Table 1: Measured synthetic smoke-test outputs. A mass residual is the volume change minus integrated SMB and moving-boundary flux, normalized by old volume.

At the final synthetic timestep, advancing-minus-control gate thickness is 0.0604 m and gate speed is -13.700 m/year. The signs indicate short-term thickening and slowdown in this particular test. They do not demonstrate a stable 68-year historical response or reproduce the magnitude of Figure 13. The paired validator confirmed equal input hashes, equal initial nodal states, equal post-spinup sampled fields, stationary control front, prescribed advancing front, finite positive thickness

and small mass residuals. Two independent unit tests checked chronology and mixed-law limits. Additional Firedrake checks verify linear-element sampling, exact volume-preserving transfer and the potential derivative against the intended mixed-law stress.

4 Reproduction and remaining limitations

`data/README.md` records required sources, units, masks and a GeoTIFF preparation manifest. Raw and processed regional datasets are ignored by git. The notebook retains executed synthetic outputs and explicitly reports the absence of historical results. `make_figure13.py` plots actual saved gate curves and maps; it refuses incomplete historical runs and requires explicit synthetic labeling. The figure comparison subtracts each case’s post-spinup baseline before subtracting control from advancing-front change.

For a defensible EKaS comparison, prepare verified regional geometry, MAR 2021 SMB and MEaSUREs 2021 velocity; calibrate the friction and rate factor; replace approximate front offsets with mapped positions; and assess the corrected bed. Then run both full experiments, halve timestep and mesh spacing, and compare flux, thickness, velocity and mass budgets. The complete-run validator tests thickening and slowdown without forcing their signs. Lack of agreement should be documented rather than tuned away without an observational basis.

The full basin, inverted A/C fields, exact bed correction, and curved front geometry are major differences from Ua. Corridor stretching and CG1 implicit transport may add numerical diffusion. The qualitative historical response, long-run stability and quantitative agreement remain unvalidated. The next scientific step is explicit sediment thickness and bank geometry, after establishing the geometric control experiment.